



[4658] – 594

Seat No.	
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T.E. (Computer) (Semester – I) Examination, 2014
OPERATING SYSTEM DESIGNS
(2012 Course)

Time : 3 Hours

Max. Marks : 70

1. A) Explain the race condition in assigning inodes. 4
B) Explain in detail monolithic kernel and micro kernel. 6
OR
2. A) Elaborate on the following algorithms in brief (any two). 8
1) iget 2) ifree 3) namei 4) getblk.
B) What is system call ? 2
3. A) Describe the structure of a regular file with proper diagrammatic representation. 6
B) What is TLB ? Why it is used ? 4
OR
4. A) Explain the concept of a region. What does the region table entry consists of ? 4
B) If the Page address stream is {2, 3, 2, 1, 5, 2, 4, 5, 3, 2, 5, 2}, and frame size is 3. Identify the page faults occurred using FIFO, LRU. 6
5. A) Explain IPC mechanisms used in System V. 8
B) What is socket ? Write and explain an algorithm to transfer data between two computers using socket. 8
OR
6. A) What is semaphore ? Provide solution to producer-consumer problem using semaphore. 8
B) Explain in detail shared memory and message passing along with their system calls. 8
7. A) What is AWK scripting ? Write an AWK script to print squares of numbers from 1 to 10. 8
B) Explain in detail how to make USB bootable with any open source tool/utility ? 8

OR

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| 8. A) What is secure boot ? State the difference between BIOS and UEFI. | 8 |
| B) What is make utility ? Explain it with example. Consider your own makefile. | 8 |
| 9. A) Draw and explain Android OS architecture. | 6 |
| B) Explain static table-driven approach for real time scheduling. | 6 |
| C) Write short notes on : 1) Fail soft operation 2) Frame of references. | 6 |

OR

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| 10. A) Explain the design issues of multiprocessor scheduling. | 6 |
| B) Compare hard, soft and firm real time systems. | 6 |
| C) Write a note on handheld devices. List various OS used for handheld devices. | 6 |
